

Completion Report: oic-stock-alert

Date: 2026-05-26 | Level: Starter | Scope: Alpha Vantage live provider + zero-trust .env

Repository: <https://github.com/ai-ml-kiosk/test-oic-stock-api>

1. Summary

1.1 Feature Overview

The OIC Stock Alert API is a dependency-free Python HTTP service for Oracle Integration Cloud orchestration. It evaluates stock alert rules, returns OIC-friendly JSON routing metadata, and preserves a stable Switch branch contract:

`response.oic.switchBranch`

The current 1.1 cycle extends the original mock-provider MVP with:

- Alpha Vantage GLOBAL_QUOTE live quote provider.
- Local untracked .env parsing for sensitive provider configuration.
- .env.example placeholder template.
- Alpha Vantage response normalization into the internal Quote model.
- Provider mapping for empty quote, rate-limit note, invalid-key information, malformed payload, and timeout outcomes.
- Fixture-based tests that avoid live network access and real API keys.

Supported OIC branch values remain:

- ALERT_TRIGGERED
- NO_ALERT
- INPUT_ERROR
- QUOTE_UNAVAILABLE
- PROVIDER_TIMEOUT
- SYSTEM_ERROR

1.2 Final Match Rate

96% (Target: 90%)

The Check phase found 49 implemented or substantially matched items out of 51 design items. The remaining gaps are low-risk hardening items around API-level live-provider fixture tests and OpenAPI/schema live-mode examples.

2. Related Documents

Document | Path

Plan | `docs/01-plan/features/oic-stock-alert.plan.md`

Design | `docs/02-design/features/oic-stock-alert.design.md`

HLD | `docs/02-design/features/oic-stock-alert.hld.md`

LLD | `docs/02-design/features/oic-stock-alert.lld.md`

Do notes | `docs/02-design/features/oic-stock-alert.do.md`

Gap analysis | `docs/03-analysis/oic-stock-alert.analysis.md`

Report markdown | `docs/04-report/oic-stock-alert.report.md`

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HLD PDF | [design/oic-stock-alert-hld.pdf](#)

LLD PDF | [design/oic-stock-alert-lld.pdf](#)

3. Completed Items

- [x] Updated plan scope for live market data orchestration while preserving mock mode.
- [x] Added FR-009 for untracked .env zero-trust credential parsing.
- [x] Added FR-010 for Alpha Vantage GLOBAL_QUOTE outbound mapping.
- [x] Updated design, HLD, and LLD for Alpha Vantage and .env behavior.
- [x] Implemented GET /health.
- [x] Implemented POST /v1/alerts/evaluate.
- [x] Preserved request normalization for request ID, symbol, currency, provider mode, and OIC tracking ID.
- [x] Preserved validation for symbols, rule counts, duplicate rule IDs, metrics, operators, thresholds, severity, message length, and JSON body size.
- [x] Preserved deterministic mock quote provider and OIC branch fixtures.
- [x] Implemented Alpha Vantage live provider module.
- [x] Implemented Alpha Vantage GLOBAL_QUOTE query mapping.
- [x] Implemented Alpha Vantage response normalization for price, change, change percent, high, low, volume, as-of date, currency, and provider identity.
- [x] Implemented outbound timeout handling using QUOTE_PROVIDER_TIMEOUT_MS.
- [x] Implemented .env parser and settings defaults.
- [x] Preserved process environment precedence over .env.
- [x] Added .env.example with placeholder values only.
- [x] Kept .env and .env* ignored by Git.
- [x] Implemented provider mappings for Alpha Vantage empty quote, rate-limit note, invalid-key information, timeout, invalid JSON, and malformed quote payloads.
- [x] Added Alpha Vantage provider tests.
- [x] Added .env parser tests.
- [x] Kept unit, API contract, and artifact tests passing.

4. Deviations From Design

Deviation | Impact | Decision

API-contract-level live provider fixture tests are not yet implemented. | Low. Provider-level fixture tests cover live behavior without network or real keys. | Add endpoint-level fixture tests in a hardening pass if needed.

OpenAPI and JSON Schema artifacts do not yet include live-mode examples. | Low. Runtime request/response contract remains unchanged. | Extend artifacts during the next extraction/report cycle.

Health endpoint still reports runtime version 1.0.0. | Low. Implementation behavior is correct; version string is release metadata. | Bump runtime version during release packaging.

Known provider client errors map to provider branches rather than SYSTEM_ERROR. | Low. Stable OIC exception branches are preserved. | Keep provider-specific classifications for actionable OIC routing.

5. Metrics

Metric | Value

Final match rate | 96%

Target match rate | 90%

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PDCA iterations | 2
Automated tests | 26
Test command | python3 -m unittest discover -s tests
Runtime dependencies | 0 third-party packages
Live provider | Alpha Vantage GLOBAL_QUOTE
Secret files committed | None
.env.example committed | Yes, placeholder only
HLD PDF pages | 5
LLD PDF pages | 9

6. Quality Metrics

Area | Result
Unit tests | Passing
API contract tests | Passing
Artifact tests | Passing
Alpha Vantage provider tests | Passing
.env parser tests | Passing
JSON artifact parse checks | Passing
OIC Switch branch coverage | Complete for designed branches
Secret handling | .env ignored; .env.example placeholder only
Network-free test suite | Passing without live Alpha Vantage calls

7. Deliverables

Deliverable | Path
Alpha Vantage provider | oic_stock_alert/alpha_vantage_provider.py
.env loader | oic_stock_alert/env_loader.py
Provider facade | oic_stock_alert/quote_provider.py
Runtime config | oic_stock_alert/config.py
.env.example | .env.example
Alpha Vantage tests | tests/test_alpha_vantage_provider.py
.env tests | tests/test_env_loader.py
OpenAPI contract | artifacts/openapi/oic-stock-alert.openapi.json
Request JSON Schema | artifacts/json-schema/alert-evaluation-request.schema.json
Response JSON Schema | artifacts/json-schema/alert-evaluation-response.schema.json
OIC Switch guide | artifacts/oic/oic-switch-branches.md

8. Lessons Learned

1. Keeping the provider boundary explicit made it straightforward to add Alpha Vantage without changing OIC payloads.
2. .env loading should stay small and non-executing; shell-style evaluation would create unnecessary secret-handling risk.
3. Provider tests should use fixtures by default so CI never depends on live market data quota or credentials.
4. OIC routing benefits from stable branch enums even when upstream provider behavior varies.
5. Runtime secrets and observability need to be designed together so provider errors remain useful without leaking API keys.

9. Follow-up Items

- [] Add API-contract-level live-provider fixture tests.
- [] Extend OpenAPI and JSON Schema examples with live-mode notes.
- [] Bump runtime health version from 1.0.0 to 1.1.0 during release packaging.

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- [] Consider optional Alpha Vantage entitlement configuration for premium delayed or realtime keys.
- [] Decide whether provider rate-limit responses should ever set `oic.retryRecommended=true`.

10. Final Recommendation

Proceed to archive after stakeholder review. The feature exceeds the PDCA threshold, keeps the mock path stable, adds Alpha Vantage live-market orchestration, and introduces a local zero-trust `.env` configuration layer without exposing secrets to Git.